Saleh Aldhafeeri

Aerospace Engineering
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Summary

Aerospace engineer working across GNC, systems engineering and data-driven problem solving. Industry experience in airworthiness at Saudia Technic and Aramco and in strategy at Accenture; founded Waypoint AI, an LLM routing venture.

Education

KAUST, *TIE: Technology, Innovation & Entrepreneurship* (Thuwal, Saudi Arabia) Aug 2026 – May 2027
Penn State University, *B.S. Aerospace Engineering* (University Park, PA) Jan 2023 – May 2026

Professional Experience

Strategy New Associate, *Accenture* (Full-time) May 2026 – Jul 2026

- Supported digital transformation engagements, helping clients restructure how they deliver and govern digital capabilities.
- Contributed to target operating model design: mapped current-state processes, roles and governance, identified gaps, and helped define future-state structures and decision rights.

Founder, *Waypoint AI* (Startup) May 2025 – Mar 2026

- Founded and ran a self-funded LLM routing venture solo, from first prototype to three paying clients.
- Built a router that scored each prompt and sent it to the cheapest capable model, escalating on low-confidence answers.
- Cut inference spend by ∼60% against an all-frontier baseline while keeping quality within 2% on a 1,200-prompt eval set.

Aircraft Engineer, *Saudia Technic* (Internship, Jeddah, Saudi Arabia) May 2025 – Sep 2025

- Resolved 50+ maintenance queries in AMOS against CMM/AMM/SRM technical manuals, achieving a 95% first-pass engineering order approval rate.
- Supported line and hangar operations on A320/321, B787-9/10 and B777-300 fleets across 10+ scheduled weekly checks.
- Verified post-maintenance airworthiness for 7 aircraft against GACAR, FAA and EASA requirements, maintaining weight and balance compliance prior to release.

Research Assistant, *Penn State University* (University Park, PA) Jan 2024 – May 2025

- Research across autonomous vehicles, GNC system design and aerodynamic modeling.

Aircraft Engineer, *Aramco, Aloula MRO* (Internship, Dhahran, Saudi Arabia) May 2024 – Sep 2024
Teaching Assistant, MATH 141 (Calculus II), *Penn State University* (Part-time, State College, PA) Aug 2023 – Apr 2024

Projects

Prometheus Mission I: Autonomous Satellite Refueling Aug 2025 – Dec 2025

- Led a 6-engineer team designing a 70 kg autonomous refueling payload for LEO satellites, delivering ± 2 mm position and $\pm 1°$ attitude accuracy at docking contact through EKF-based relative navigation.

Autonomous Drone Landing on a Moving Object Jan 2025 – May 2025

- Built a cascaded PID/PD quadrotor guidance system for autonomous landing on a moving vehicle; achieved centimeter-level touchdown via velocity feedforward and flight-phase logic.

Extended Kalman Filter: Lunar Navigation System Jan 2025 – May 2025

- Developed a 9-state EKF in MATLAB running at 100 Hz for lunar navigation with multi-rate IMU fusion; validated within 2σ consistency bounds.

Turbine Blade CFD: Aerodynamic Design & Analysis Aug 2024 – Dec 2024

- Modeled a parametric 3D blade in SolidWorks; reduced total-pressure loss by 12% and raised section L/D by 18% through CFD optimization.

Leadership

Board Member, *Nittany Lion Consulting Group* (University Park, PA) Jan 2024 – May 2025

- Delivered 6+ client engagements; reduced average case turnaround by 20% through standardized hypothesis-driven analysis frameworks.

Treasurer, *Penn State MENA Caucus Club* Dec 2023 – Dec 2024

- Built and launched a secure finance website for 100+ members using JavaScript, Python and PostgreSQL; managed a $10K event budget across 7 fundraisers.

Skills & Certifications

Skills: Systems Engineering, GNC, EKF / INS, Spacecraft Design, Compressible CFD, Airworthiness, CAD / FEA, MATLAB, ANSYS Fluent, SolidWorks, STK, Python, LLM Infrastructure
Certifications: Ansys STK Grand Master (L3), Ansys Fluent (2024), SolidWorks CSWE (2024), MATLAB Associate (2023)